

Energy Management System (EMS) 3D Smart Industrial Digital Twin

Complete Step-by-Step Procedure: Modeling, 3D Graphics Engineering, HUD Projection & Cloud Deployment

Platform: [Three.js / WebGL 2.0](#)

Source: [Image 1 \(Specs\)](#) & [Image 2 \(EMS Dashboard\)](#)

Repository: [binduniharika09/energy-management-system](#)

00 Executive Summary

This document outlines the complete engineering workflow undertaken to build the interactive **3D Industrial Smart Campus** and **Energy Management System (EMS)**. The project transforms raw electrical consumption telemetry and contract demand parameters from a 12-division industrial matrix ([Image 1](#)) into an interactive, real-time 3D digital twin platform ([Image 2](#)) with animated power conduits, dynamic camera controls, and analytical diagnostic drawers.

01 Phase 1: Requirements Analysis & Telemetry Extraction

The initial step involved extracting and mathematically verifying the operational parameters for all 12 industrial manufacturing units given in [Image 1](#). The total connected load across all 12 units sums exactly to **10,394 kW**, matching the live primary telemetry KPI displayed in [Image 2](#).

#	DIVISION NAME	MAX DEMAND (KVA)	CONNECTED LOAD (KW)	PF	MONTHLY CONS. (KWH)
1	PEB Structure Manufacturing	150	679	0.96	38,550
2	ROB Structure Manufacturing	100	532	0.95	10,718
3	CNG Booster compressor manufacturing	40	67	0.99	5,432
4	Fastners manufacturing unit	180	718	0.95	33,059
5	Transmission Tower manufacturing unit	250	1,523	0.95	39,709
6	Solar Structure Manufacturing	120	990	0.99	20,070
7	Scaffolding fabrication unit	98	453	0.98	25,712
8	Signboard manufacturing unit	25	134	0.99	3,426
9	Galvanising Unit	120	551	0.99	98,437
10	Small arm manufacturing unit	60	161	0.99	3,487
11	Composite unit	100	2,188	0.99	20,900
12	Shelters manufacturing unit	120	2,398	0.98	37,695

#	DIVISION NAME	MAX DEMAND (KVA)	CONNECTED LOAD (KW)	PF	MONTHLY CONS. (KWH)
Σ	Total / Campus Aggregates	1,263 kVA (1,400 Cont.)	10,394 kW	0.97 Avg	3,84,791 kWh

02

Phase 2: Tech Stack & System Architecture

Core 3D Engine: Three.js (r128)

High-performance WebGL 2.0 renderer capable of rendering multi-mesh scenes, PBR materials, soft directional shadows, and bloom post-processing at steady 60 FPS across desktop and mobile.

Navigation: OrbitControls

Constrained orbital camera system supporting smooth 360° rotational inspection, responsive zooming, and clamped polar angles to prevent subterranean camera clipping.

Animation: Tween.js

Interpolates camera positions and look-at target vectors using cubic-out easing curves for cinematic fly-to transitions when selecting industrial buildings.

Visualization: Chart.js

Lightweight canvas charting framework for rendering historical monthly consumption profiles, load fluctuations, and equipment telemetry in the unit inspection modal.

03 Phase 3: Spatial Coordinate Modeling & Data Structuring

The industrial campus follows a radial layout centered on the 33 kV substation. In `js/data.js`, each unit was assigned a radial distance (r) and an angular orientation (θ) matching the circular layout of Image 2:

Spatial Transformation Formula:

$$X = r \cdot \cos(\theta_{\text{rad}}), \quad Z = -r \cdot \sin(\theta_{\text{rad}}), \quad \text{Rotation}_Y = -\theta_{\text{rad}} + \frac{\pi}{2}$$
 This formula places each factory along the ring road and automatically orients the factory's front loading apron towards the center of the campus.

04 Phase 4: 3D Scene Engineering (`js/scene3d.js`)

1. Central 33 kV Energy Hub Substation

- Energy Reactor Core:** Modeled with an illuminated neon green cylinder (`#00e676`) housed inside a transparent refractive glass cylinder.
- Dual Gyro Rings:** Two counter-rotating holographic torus rings that spin on the X/Y axes during runtime.
- Transformer Banks:** Four 3D step-down oil transformer tanks with cooling radiators and high-voltage bushing insulators.
- Lattice Gantry Towers:** Structural lattice pylons, crossbeams, and busbar insulator strings.

2. 12 Parametric Manufacturing Units

- Industrial Warehouses:** Modeled with pitched steel roof frames, industrial cladding, roll-up loading bays, and glass clerestory windows.
- Photovoltaic Solar Arrays:** High-density photovoltaic panels mounted across the roofs, matching Image 2's clean-energy campus design.
- Distinct Structural Props:** High-tonnage gantry cranes for Units 1 & 2; gas storage tanks for Unit 3; exhaust chimneys for Units 4 & 9; prototype transmission towers for Unit 5; autoclave curing chambers for Unit 11.

3. Dynamic Glowing Energy Conduit Beams

- Constructed 3D `QuadraticBezierCurve3` splines arching upward from the central hub into each factory's step-down transformer box.
- Generated animated particle spheres that traverse the curves continuously, producing a live visual energy pulse.

4. Campus Landscaping & Perimeter Forest

- Concentric asphalt road rings (inner loop and outer highway) and 12 radial connection streets.
- Surrounding forest of 380+ procedural 3D conifers and deciduous trees with randomized green foliage shaders.

05 Phase 5: 3D-to-2D Projected UI HUD & Marker Occlusion

Floating callout cards were overlaid on the 3D scene using screen coordinate projection:

- **World-to-Screen Mapping:** In each render frame, building 3D coordinates (X, Y, Z) are projected using `vector.project(camera)` to calculate 2D viewport coordinates (x, y) .
- **Frustum Occlusion Elimination:** A vector dot product check with the camera's forward vector ($\vec{d}_{\text{cam}} \cdot \vec{d}_{\text{obj}} \leq 0$) hides labels when their building moves behind the camera viewpoint, preventing UI inversion.

06 Phase 6: Interactive Dashboard & Diagnostics (`js/app.js`)

- **Header Telemetry:** Live aggregated metrics (10,394 kW Load, 3,84,791 kWh Consumption, 0.97 Avg PF, Live Date/Clock).
- **Right Analytics Panel:** Animated live energy flow schematic ('Grid ↔ Hub ↔ Units'), environment impact ('21.35 Tonnes CO₂ Saved'), and status flags.
- **Diagnostic Inspection Drawer:** Clicking any unit flies the 3D camera to that building and opens a modal with equipment load ratings and historical consumption charts.
- **Simulation Ticker:** Simulates realistic $\pm 1.5\%$ micro-fluctuations in electrical loads to provide a living digital twin experience.

07 Phase 7: Version Control & GitHub Pages Deployment

1. Extracted the 3D module into a dedicated branch using `git subtree split --prefix=ems-3d-module -b ems-release`.
2. Pushed the branch directly to the root of the new GitHub repository:
`https://github.com/binduniharika09/energy-management-system.git`.
3. Configured GitHub Pages to serve the 3D platform live from the `main` branch root:
`https://binduniharika09.github.io/energy-management-system/`